

Connect to CONSERVE

Save on resources with this workshop on setting up a LAN, sharing your Internet connection between PCs, and setting up a mail server

Many of us have more than one PC in our homes and workplaces. Having them wired on a network has several benefits that will allow you to utilise resources more efficiently. It's really quite easy to set up a network for simple tasks such as sharing files and folders, sharing a printer or even your Internet connection. You can even set up an e-mail server that automatically downloads mail for everyone and distributes it over the network. You will be surprised at the cost savings when you decide to share your resources.

Putting the pieces together

Now that you've got all the pieces for your network in place (*Check out the 'What you need' box for more details*), let's put them together.

First, identify the PC that will connect directly to the Internet. In case this PC has a cable or DSL connection you will need to install two network cards. If you're using a dialup modem you just need one network card per machine. Make sure that your PC has free PCI slots to plug in the cards.

Next, you need to connect the cables. Arrange the cables neatly such that they don't get cut between doors and no one trips over them. Now connect the two PCs with the crossover cable (or use a hub if you're connecting more than two computers).

Selecting the right driver

If you are using Windows 2000 or XP, the network cards should be detected automatically. If they aren't, you will

have to manually install the drivers using the installation disk that came along with the card. When you have done this, you will be asked to enter the computer and domain name. For example, let's name our domain 'MyNet' and the PCs as 'Home1' and 'Home2'. Home1 is the PC that is connected to the Net (the host) and Home2 is the client machine.

Setup your network adapter

Now that you've plugged in the network cards, you need to configure your network. Right-click on the 'Network Neighbourhood' desktop icon on Home1 and select 'Properties'. 'TCP/IP' should be the default protocol for your network. Ensure that you've installed 'Client for Microsoft Networks' and 'File and Printer sharing for Microsoft Networks'. Now select the 'TCP/IP' component and click the 'Properties' button. This brings up the 'TCP/IP Properties' dialog box. Click on the IP (Internet Protocol) Address tab and enter the IP as 192.168.0.1. In Home2, you need to enter the IP as 192.168.0.2. Also under the 'Gateway' tab of Home2, enter the IP of Home1, i.e. 192.168.0.1. The subnet masks for both the machines should be 255.255.255.0.

Those with cable or DSL connections will need to enter additional details such as the IP address assigned by the service provider, Gateway IP, and DNS server details for the network card (in Home1) to which the cable or DSL modem is attached. You can get these details by contacting your service provider. The client machine's settings remain unchanged (the same as mentioned above).

NATing it up

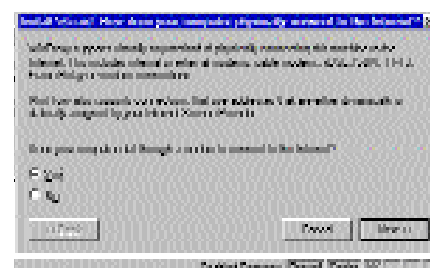
NAT (Network Address Translation) allows LANs to use two sets of IP addresses (internal and external) to access the Internet. An IP is your PC's address on the network (LAN or Internet). The external IP is the static or dynamic IP that your ISP assigns to you. Internal IPs are used by NAT so that other PCs on the network can access the Internet even though they aren't directly connected to the Internet. Whenever computers want to access the Net, NAT converts the internal IPs so that they can be used for accessing the Internet.

For our network, we'll be using a NAT-based proxy server. Proxy servers allow the sharing of a single Internet connection over a network.

Enter WinProxy

WinProxy is a NAT based shareware proxy server with support for internal and external modems as well as cable and DSL connections.

The configuration wizard will automatically start up when you launch WinProxy for the first time. You will need to specify whether your computer uses a modem for Internet access. Select 'Yes' if



Select your kind of Internet connection in WinProxy